

	$\mathcal{K}_0^{\#1}$	$\mathcal{K}_0^{\#1} \alpha\beta X$						
$\mathcal{K}_0^{\#1+}$	0	0						
$\mathcal{K}_0^{\#1+}\alpha\beta X$	0	$\gamma_1 k^2 + 3 \binom{(2)}{k_1}$	$\mathcal{K}_1^{\#1} \alpha\beta$	$\mathcal{K}_1^{\#2} \alpha\beta$	$\mathcal{K}_1^{\#1} \alpha$	$\mathcal{K}_1^{\#2} \alpha$		
	$\mathcal{K}_1^{\#1+}\alpha\beta$	$\gamma_2 k^2 + 2 \binom{(2)}{k_1}$	$\frac{-\gamma_3 k^2 + 4 \binom{(2)}{k_1}}{2 \sqrt{2}}$	0	0			
	$\mathcal{K}_1^{\#2+}\alpha\beta$	$\frac{-\gamma_3 k^2 + 4 \binom{(2)}{k_1}}{2 \sqrt{2}}$	$\frac{1}{\sqrt{2}} (\gamma_4 k^2 + 2 \binom{(2)}{k_1})$	0	0			
	$\mathcal{K}_1^{\#1+}\alpha$	0	0	$\frac{\gamma_3 k^2}{2}$	$\frac{\gamma_3 k^2}{\sqrt{2}}$			
	$\mathcal{K}_1^{\#2+}\alpha$	0	0	$\frac{\gamma_5 k^2}{\sqrt{2}}$	$\gamma_5 k^2$			
			$\mathcal{K}_2^{\#1+}\alpha\beta$	$\mathcal{K}_2^{\#1} \alpha\beta X$				
			$\mathcal{K}_{2+}^{\#1+}\alpha\beta$	0	0			
			$\mathcal{K}_2^{\#1+}\alpha\beta X$	0	$\frac{\gamma_6 k^2}{2}$			

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^+ \beta X}^{\#1}$				
$\mathcal{T}_{0^+}^{\#1} \dagger$	0	0				
$\mathcal{T}_{0^+}^{\#1} + \alpha \beta X$	0	$\frac{1}{\text{Det}(0)}$	$\mathcal{T}_{1^+ \alpha \beta}^{\#1}$	$\mathcal{T}_{1^+ \alpha \beta}^{\#2}$	$\mathcal{T}_{1^+ \alpha}^{\#1}$	$\mathcal{T}_{1^+ \alpha}^{\#2}$
$\mathcal{T}_{1^+}^{\#1} + \alpha \beta$	$\frac{\gamma_4 k^2 + 2}{2 \text{Det}(1^+)}$	$\frac{\gamma_4 k^2 + 2}{2 \text{Det}(1^+)}$	$\frac{\gamma_3 k^2 + 4}{2 \sqrt{2} \text{Det}(1^+)}$	0	0	
$\mathcal{T}_{1^+}^{\#2} + \alpha \beta$	$\frac{\gamma_3 k^2 + 4}{2 \sqrt{2} \text{Det}(1^+)}$	$\frac{\gamma_2 k^2 + 2}{\text{Det}(1^+)}$	0	0		
$\mathcal{T}_{1^+}^{\#1} + \alpha$	0	0	$\frac{1}{3 \text{Det}(1^+)}$	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$		
$\mathcal{T}_{1^+}^{\#2} + \alpha$	0	0	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\frac{2}{3 \text{Det}(1^+)}$	$\mathcal{T}_{2^+ \alpha \beta}^{\#1}$	$\mathcal{T}_{2^+ \alpha \beta X}^{\#1}$
			$\mathcal{T}_{2^+}^{\#1} + \alpha \beta$	0	0	
			$\mathcal{T}_{2^+}^{\#1} + \alpha \beta X$	0	$\frac{1}{\text{Det}(2)}$	

$$\begin{aligned} Y_1 &= \kappa_{15}^{(4)} - \kappa_{16}^{(4)} \& Y_2 = 2 \kappa_{15}^{(4)} - \kappa_3^{(4)} \& Y_3 = 2 \kappa_{16}^{(4)} - \kappa_4^{(4)} \& Y_4 = 9 \kappa_1^{(4)} + 8 \kappa_{15}^{(4)} + 3 \kappa_{16}^{(4)} + \kappa_3^{(4)} + \kappa_4^{(4)} \& \\ Y_5 &= 2 \left(\kappa_1^{(4)} + \kappa_{15}^{(4)} \right) + \kappa_{16}^{(4)} \& Y_6 = 2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)} \& \text{Det}(0^+) = k^2 \left(\kappa_{15}^{(4)} - \kappa_{16}^{(4)} \right) + 3 \kappa_1^{(2)} \& \\ \text{Det}(1^+) &= \frac{1}{8} k^4 \left(72 \kappa_1^{(4)} \kappa_{15}^{(4)} + 64 \kappa_{15}^{(4)2} + 24 \kappa_{15}^{(4)} \kappa_{16}^{(4)} - 4 \kappa_{16}^{(4)2} - 12 \left(3 \kappa_1^{(4)} + 2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)} \right) \kappa_3^{(4)} - 4 \kappa_3^{(4)2} + \right. \\ &\quad \left. 4 \left(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)} - \kappa_3^{(4)} \right) \kappa_4^{(4)} - \kappa_4^{(4)2} \right) + k^2 \left(9 \kappa_1^{(4)} + 5 \left(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)} \right) \kappa_1^{(2)} \& \text{Det}(2^+) = \frac{1}{2} k^2 \left(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)} \right) \end{aligned}$$

$$\begin{aligned} & \text{Lagrangian} \\ & \frac{(2)}{\kappa_1} \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - 2 \frac{(2)}{\kappa_1} \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \frac{(4)}{\kappa_1} \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^{\alpha\beta}_{\alpha\beta} - \frac{(4)}{\kappa_1} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\beta \mathcal{K}^{\alpha\beta}_{\alpha\beta} + \\ & \frac{(4)}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\alpha\chi} + \frac{(4)}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} + 2 \frac{(4)}{\kappa_{15}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\chi\alpha} + \frac{(4)}{\kappa_{16}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\chi\alpha} . \\ & \frac{(4)}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\chi\alpha} + 6 \frac{(4)}{\kappa_1} \partial^\chi \mathcal{K}^{\alpha\beta}_{\alpha\beta} \partial_\delta \mathcal{K}^{\delta}_{\chi\beta} + 4 \frac{(4)}{\kappa_{15}} \partial^\chi \mathcal{K}^{\alpha\beta}_{\alpha\beta} \partial_\delta \mathcal{K}^{\delta}_{\chi\beta} + 2 \frac{(4)}{\kappa_{16}} \partial^\chi \mathcal{K}^{\alpha\beta}_{\alpha\beta} \partial_\delta \mathcal{K}^{\delta}_{\chi\beta} + \\ & \frac{9}{2} \frac{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} + 3 \frac{(4)}{\kappa_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} + \frac{3}{2} \frac{(4)}{\kappa_{16}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} + \\ & \frac{1}{2} \frac{(4)}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} + \frac{1}{2} \frac{(4)}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} + \kappa_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \kappa_{16} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):		$\mathcal{K}^{a\beta\chi} \mathcal{J}_{a\beta\chi}$
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_0^{\#1} = 0$	1	$\partial_\beta \mathcal{J}^\alpha{}_\alpha = 0$
$\mathcal{J}_1^{\#1\alpha} = \mathcal{J}_1^{\#2\alpha}$	3	$\partial_\lambda \partial^\alpha \mathcal{J}^\beta{}_\beta{}^\chi + \partial_\lambda \partial^\chi \mathcal{J}^{\beta\alpha}{}_\beta = 0$
$\mathcal{J}_2^{\#1\alpha\beta} = 0$	5	$3 \partial_\delta \partial_\lambda \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\lambda \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{a\beta} \partial_c \partial_c \partial_\delta \mathcal{J}^{\chi\chi}{}_\lambda{}^\delta = 2 \partial_\delta \partial^\delta \partial^\alpha \mathcal{J}^{\chi\chi}{}_\lambda{}^\delta + 3 (\partial_\delta \partial^\delta \partial_\lambda \mathcal{J}^{a\beta\chi} + \partial_\delta \partial^\delta \partial_\lambda \mathcal{J}^{\beta\alpha\chi})$

Total # constraint(s):	9		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{21 \binom{(4)}{1}^2 + 18 \binom{(4)}{1} (2 \binom{(4)}{15} + \binom{(4)}{16}) + 4 (2 \binom{(4)}{15} + \binom{(4)}{16})^2}{(2 \binom{(4)}{15} + \binom{(4)}{16}) (2 \binom{(4)}{1} + 2 \binom{(4)}{15} + \binom{(4)}{16}) (9 \binom{(4)}{1} + 5 (2 \binom{(4)}{15} + \binom{(4)}{16}))}$

$$\begin{aligned}
& (1944 \binom{(4)}{K_1}^3 \binom{(2)}{K_1} + 6264 \binom{(4)}{K_1}^2 \binom{(4)}{K_{15}} \binom{(2)}{K_1} + 6720 \binom{(4)}{K_1} \binom{(4)}{K_{15}}^2 \binom{(2)}{K_1} + 2400 \binom{(4)}{K_{15}}^3 \binom{(2)}{K_1} + 3132 \binom{(4)}{K_1}^2 \binom{(4)}{K_{16}} \binom{(2)}{K_1} + \\
& 6720 \binom{(4)}{K_1} \binom{(4)}{K_{15}} \binom{(4)}{K_{16}} \binom{(2)}{K_1} + 3600 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_{16}} \binom{(2)}{K_1} + 1680 \binom{(4)}{K_1} \binom{(4)}{K_{16}}^2 \binom{(2)}{K_1} + 1800 \binom{(4)}{K_{15}} \binom{(4)}{K_{16}} \binom{(2)}{K_1} + \\
& 300 \binom{(4)}{K_{16}}^3 \binom{(2)}{K_1} + 2592 \binom{(4)}{K_1}^3 \binom{(4)}{K_{15}} p^2 + 5760 \binom{(4)}{K_1}^2 \binom{(4)}{K_{15}}^2 p^2 + 4064 \binom{(4)}{K_1} \binom{(4)}{K_{15}}^3 p^2 + 896 \binom{(4)}{K_{15}}^4 p^2 + \\
& \binom{(4)}{K_1}^3 \binom{(4)}{K_{16}}) = \binom{(4)}{K_1}^2 \binom{(4)}{K_{15}} \binom{(4)}{K_{16}}) = \binom{(4)}{K_1} \binom{(4)}{K_{15}}^2 \binom{(4)}{K_{16}}) = \binom{(4)}{K_{15}}^3 \binom{(4)}{K_{16}}).
\end{aligned}$$
[illegible]



$$\begin{aligned}
& 144 \kappa_{15}^{(4)} \kappa_3^{(4)} \kappa_4^{(4)} p^2 + 72 \kappa_1^{(4)} \kappa_{16}^{(4)} \kappa_3^{(4)} \kappa_4^{(4)} p^2 + 144 \kappa_{15}^{(4)} \kappa_{16}^{(4)} \kappa_3^{(4)} \kappa_4^{(4)} p^2 + 36 \kappa_{16}^{(4)} \kappa_3^{(4)} \kappa_4^{(4)} p^2 + \\
& 36 \kappa_1^{(4)} \kappa_{15}^{(4)} \kappa_4^{(4)} p^2 + 36 \kappa_{15}^{(4)} \kappa_4^{(4)} p^2 + 18 \kappa_1^{(4)} \kappa_{16}^{(4)} \kappa_4^{(4)} p^2 + 36 \kappa_{15}^{(4)} \kappa_{16}^{(4)} \kappa_4^{(4)} p^2 + 9 \kappa_{16}^{(4)} \kappa_4^{(4)} p^2 - \\
& \sqrt{((2 \kappa_1^{(4)} + 2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)})^2 (104976 \kappa_1^{(4)}{}^4 (72 \kappa_1^{(2)}{}^2 + 30 (2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)})^2 \kappa_1^{(2)} p^2 + 13 (2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)})^2 p^4) + \\
& 11664 \kappa_1^{(4)}{}^3 (2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (1494 \kappa_1^{(2)}{}^2 + 6 (164 \kappa_{15}^{(4)} + 82 \kappa_{16}^{(4)} + 18 \kappa_3^{(4)} + 9 \kappa_4^{(4)})^2 \kappa_1^{(2)} p^2 + \\
& 5 (2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (68 \kappa_{15}^{(4)} + 34 \kappa_{16}^{(4)} + 18 \kappa_3^{(4)} + 9 \kappa_4^{(4)}) p^4) + 1944 \kappa_1^{(4)}{}^2 (2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) \\
& (9 (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 \kappa_1^{(2)} p^2 + 4256 \kappa_{15}^{(4)}{}^3 p^4 + 532 \kappa_{16}^{(4)}{}^3 p^4 + 6 \kappa_{16}^{(4)}{}^2 (326 \kappa_1^{(2)} p^2 + 53 (2 \kappa_3^{(4)} + \kappa_4^{(4)}) p^4) + \\
& 3 \kappa_{16}^{(4)} (2598 \kappa_1^{(2)}{}^2 + 168 (2 \kappa_3^{(4)} + \kappa_4^{(4)}) \kappa_1^{(2)} p^2 + 11 (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 p^4) + 24 \kappa_{15}^{(4)}{}^2 \\
& (326 \kappa_1^{(2)} p^2 + (266 \kappa_{16}^{(4)} + 53 (2 \kappa_3^{(4)} + \kappa_4^{(4)}) p^4) + 6 \kappa_{15}^{(4)} (2598 \kappa_1^{(2)}{}^2 + 8 (163 \kappa_{16}^{(4)} + 21 (2 \kappa_3^{(4)} + \kappa_4^{(4)}) \kappa_1^{(2)} p^2 + \\
& (326 \kappa_1^{(2)} p^2 + (266 \kappa_{16}^{(4)} + 53 (2 \kappa_3^{(4)} + \kappa_4^{(4)}) p^4) + 6 \kappa_{15}^{(4)} (2598 \kappa_1^{(2)}{}^2 + 8 (163 \kappa_{16}^{(4)} + 21 (2 \kappa_3^{(4)} + \kappa_4^{(4)}) \kappa_1^{(2)} p^2 +
\end{aligned}$$

$$\begin{aligned}
& (532 \kappa_{16}^{(4)} + 212 \kappa_{16}^{(2)} (2 \kappa_3 + \kappa_4) + 11 (2 \kappa_3 + \kappa_4)^2 p^4) + 36 \kappa_1^{(2)} (2 \kappa_{15} + \kappa_{16})^2 \\
& (50176 \kappa_{15}^{(4)} p^4 + 6272 \kappa_{16}^{(4)} p^4 + 27 (2 \kappa_3 + \kappa_4)^2 p^2 (20 \kappa_1^{(2)} + 3 (2 \kappa_3 + \kappa_4) p^2) + 24 \kappa_{16}^{(2)} \\
& (1240 \kappa_1^{(2)} p^2 + 273 (2 \kappa_3 + \kappa_4) p^4) + 18 \kappa_{16}^{(4)} (9080 \kappa_1^{(2)} + 780 (2 \kappa_3 + \kappa_4) \kappa_1^{(2)} p^2 + 83 (2 \kappa_3 + \kappa_4)^2 p^4) + \\
& 96 \kappa_{15}^{(4)} (1240 \kappa_1^{(2)} p^2 + 7 (112 \kappa_{16}^{(4)} + 78 \kappa_3 + 39 \kappa_4) p^4) + 12 \kappa_{15}^{(2)} (27240 \kappa_1^{(2)} + \\
& 20 (496 \kappa_{16}^{(4)} + 117 (2 \kappa_3 + \kappa_4)) \kappa_1^{(2)} p^2 + (3136 \kappa_{16}^{(4)} + 2184 \kappa_{16}^{(2)} (2 \kappa_3 + \kappa_4) + 249 (2 \kappa_3 + \kappa_4)^2 p^4)) + \\
& (2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)})^2 (280576 \kappa_{15}^{(4)} p^4 + 17536 \kappa_{16}^{(4)} p^4 + 81 (2 \kappa_3 + \kappa_4)^4 p^4 + \\
& 108 \kappa_{16}^{(4)} (2 \kappa_3 + \kappa_4)^2 p^2 (50 \kappa_1^{(2)} + 9 (2 \kappa_3 + \kappa_4) p^2) + 384 \kappa_{16}^{(2)} (275 \kappa_1^{(2)} p^2 + 72 (2 \kappa_3 + \kappa_4) p^4) + \\
& 72 \kappa_{16}^{(4)} (11950 \kappa_1^{(2)} + 1200 (2 \kappa_3 + \kappa_4) \kappa_1^{(2)} p^2 + 167 (2 \kappa_3 + \kappa_4)^2 p^4) + \\
& (4 \kappa_3^3 + 3 \kappa_3^2 \kappa_4 + 3 \kappa_3 \kappa_4^2 + \kappa_4^3) p^4 + 2 \kappa_3 \kappa_4 (2 \kappa_3 + \kappa_4) p^2 + \kappa_3 \kappa_4 (2 \kappa_3 + \kappa_4)^2 p^2
\end{aligned}$$

			$ \begin{aligned} & 1024 \kappa_{15}^{(2)} (825 \kappa_{16}^{(2)} p^2 + 4 (137 \kappa_{16}^{(2)} + 54 (2 \kappa_3^{(4)} + \kappa_4^{(4)}) p^4) + 96 \kappa_{15}^{(4)} (35850 \kappa_1^{(4)} + \\ & 1200 (11 \kappa_{16}^{(4)} + 6 \kappa_3^{(4)} + 3 \kappa_4^{(4)}) \kappa_1^{(2)} p^2 + (4384 \kappa_{16}^{(4)} + 3456 \kappa_{16}^{(4)} (2 \kappa_3^{(4)} + \kappa_4^{(4)}) + 501 (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2) p^4) + \\ & 8 \kappa_{15}^{(4)} (17536 \kappa_{16}^{(4)} p^4 + 27 (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 p^2 (50 \kappa_1^{(2)} + 9 (2 \kappa_3^{(4)} + \kappa_4^{(4)}) p^2) + 288 \kappa_{16}^{(4)} (275 \kappa_1^{(2)} p^2 + \\ & 72 (2 \kappa_3^{(4)} + \kappa_4^{(4)}) p^4) + 36 \kappa_{16}^{(4)} (11950 \kappa_1^{(2)} + 1200 (2 \kappa_3^{(4)} + \kappa_4^{(4)}) \kappa_1^{(2)} p^2 + 167 (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 p^4) + \dots) / \\ & ((2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (2 \kappa_1^{(4)} + 2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (9 \kappa_1^{(4)} + 5 (2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)})^2 \kappa_1^{(2)})) \end{aligned} $
			$ \begin{aligned} & (1944 \kappa_1^{(4)} \kappa_{15}^{(3)} \kappa_1^{(2)} + 6264 \kappa_1^{(4)} \kappa_{15}^{(2)} \kappa_1^{(2)} + 6720 \kappa_1^{(4)} \kappa_{15}^{(2)} \kappa_1^{(2)} + 2400 \kappa_{15}^{(4)} \kappa_1^{(3)} \kappa_1^{(2)} + 3132 \kappa_1^{(4)} \kappa_{16}^{(2)} \kappa_1^{(2)} + \\ & 6720 \kappa_1^{(4)} \kappa_{15}^{(4)} \kappa_{16}^{(2)} \kappa_1^{(2)} + 3600 \kappa_{15}^{(4)} \kappa_{16}^{(2)} \kappa_1^{(2)} + 1680 \kappa_1^{(4)} \kappa_{16}^{(2)} \kappa_1^{(2)} + 1800 \kappa_{15}^{(4)} \kappa_{16}^{(2)} \kappa_1^{(2)} + \dots) \end{aligned} $

$$\begin{aligned}
& 300 K_{16}^{(4)} K_{11}^{(4)} + 2592 K_{11}^{(4)} K_{15}^{(4)} p^2 + 5760 K_{11}^{(4)} K_{15}^{(4)} p^2 + 4064 K_{11}^{(4)} K_{15}^{(4)} p^2 + 896 K_{15}^{(4)} p^2 + \\
& 1296 K_{11}^{(4)} K_{16}^{(4)} p^2 + 5760 K_{11}^{(4)} K_{15}^{(4)} K_{16}^{(4)} p^2 + 6096 K_{11}^{(4)} K_{15}^{(4)} K_{16}^{(4)} p^2 + 1792 K_{15}^{(4)} K_{16}^{(4)} p^2 + \\
& 1440 K_{11}^{(4)} K_{16}^{(4)} p^2 + 3048 K_{11}^{(4)} K_{15}^{(4)} K_{16}^{(4)} p^2 + 1344 K_{15}^{(4)} K_{16}^{(4)} p^2 + 508 K_{11}^{(4)} K_{16}^{(4)} p^2 + \\
& 448 K_{15}^{(4)} K_{16}^{(4)} p^2 + 56 K_{16}^{(4)} p^2 + 1296 K_{11}^{(4)} K_{15}^{(4)} K_3^{(4)} p^2 + 2160 K_{11}^{(4)} K_{15}^{(4)} K_3^{(4)} p^2 + 864 K_{15}^{(4)} K_3^{(4)} p^2 + \\
& 648 K_{11}^{(4)} K_{16}^{(4)} K_3^{(4)} p^2 + 2160 K_{11}^{(4)} K_{15}^{(4)} K_{16}^{(4)} K_3^{(4)} p^2 + 1296 K_{15}^{(4)} K_{16}^{(4)} K_3^{(4)} p^2 + 540 K_{11}^{(4)} K_{16}^{(4)} K_3^{(4)} p^2 + \\
& 648 K_{15}^{(4)} K_{16}^{(4)} K_3^{(4)} p^2 + 108 K_{16}^{(4)} K_3^{(4)} p^2 + 144 K_{11}^{(4)} K_{15}^{(4)} K_3^{(4)} p^2 + 144 K_{15}^{(4)} K_3^{(4)} p^2 + 72 K_{11}^{(4)} K_{16}^{(4)} K_3^{(4)} p^2 + \\
& 144 K_{15}^{(4)} K_{16}^{(4)} K_3^{(4)} p^2 + 36 K_{16}^{(4)} K_3^{(4)} p^2 + 648 K_{11}^{(4)} K_{15}^{(4)} K_4^{(4)} p^2 + 1080 K_{11}^{(4)} K_{15}^{(4)} K_4^{(4)} p^2 + \\
& 432 K_{15}^{(4)} K_4^{(4)} p^2 + 324 K_{11}^{(4)} K_{16}^{(4)} K_4^{(4)} p^2 + 1080 K_{11}^{(4)} K_{15}^{(4)} K_{16}^{(4)} K_4^{(4)} p^2 + 648 K_{15}^{(4)} K_{16}^{(4)} K_4^{(4)} p^2 + \\
& 432 K_{11}^{(4)} K_{16}^{(4)} K_4^{(4)} p^2 + 324 K_{16}^{(4)} K_4^{(4)} p^2 + 1080 K_{11}^{(4)} K_{15}^{(4)} K_{16}^{(4)} K_4^{(4)} p^2 + 648 K_{15}^{(4)} K_{16}^{(4)} K_4^{(4)} p^2 +
\end{aligned}$$

	1	0	$ \begin{aligned} & 270 \kappa_1^4 \kappa_{16}^4 \kappa_4 p^2 + 324 \kappa_{15}^4 \kappa_{16}^4 \kappa_4 p^2 + 54 \kappa_{16}^4 \kappa_4 p^2 + 144 \kappa_1^4 \kappa_{15}^4 \kappa_3^4 \kappa_4 p^2 + \\ & 144 \kappa_{15}^4 \kappa_3^4 \kappa_4 p^2 + 72 \kappa_1^4 \kappa_{16}^4 \kappa_3^4 \kappa_4 p^2 + 144 \kappa_{15}^4 \kappa_{16}^4 \kappa_3^4 \kappa_4 p^2 + 36 \kappa_{16}^4 \kappa_3^4 \kappa_4 p^2 + \\ & 36 \kappa_1^4 \kappa_{15}^4 \kappa_4^2 p^2 + 36 \kappa_{15}^4 \kappa_4^2 p^2 + 18 \kappa_1^4 \kappa_{16}^4 \kappa_4^2 p^2 + 36 \kappa_{15}^4 \kappa_{16}^4 \kappa_4^2 p^2 + 9 \kappa_{16}^4 \kappa_4^2 p^2 + \\ & \sqrt{(2 \kappa_1^4 + 2 \kappa_{15}^4 + \kappa_{16}^4)^2 (104976 \kappa_1^4 (72 \kappa_1^4 + 30 (2 \kappa_{15}^4 + \kappa_{16}^4)) \kappa_1^2 p^2 + 13 (2 \kappa_{15}^4 + \kappa_{16}^4)^2 p^4) +} \\ & 11664 \kappa_1^3 (2 \kappa_{15}^4 + \kappa_{16}^4) (1494 \kappa_1^2 (2 \kappa_{15}^4 + \kappa_{16}^4) \kappa_1 p^2 + 18 \kappa_3^4 + 9 \kappa_4^4) \kappa_1^2 p^2 + \\ & 5 (2 \kappa_{15}^4 + \kappa_{16}^4) (68 \kappa_{15}^4 + 34 \kappa_{16}^4 + 18 \kappa_3^4 + 9 \kappa_4^4) p^4 + 1944 \kappa_1^2 (2 \kappa_{15}^4 + \kappa_{16}^4) \\ & (9 (2 \kappa_3^4 + \kappa_4^4)^2 \kappa_1 p^2 + 4256 \kappa_{15}^3 p^4 + 532 \kappa_{16}^3 p^4 + 6 \kappa_{16}^2 (326 \kappa_1 p^2 + 53 (2 \kappa_3^4 + \kappa_4^4) p^4) + \\ & 3 \kappa_{16}^4 (2598 \kappa_1^2 p^2 + 168 (2 \kappa_3^4 + \kappa_4^4) \kappa_1 p^2 + 11 (2 \kappa_3^4 + \kappa_4^4)^2 p^4) + 24 \kappa_{15}^4 \end{aligned} $
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$$\begin{aligned}
& (326 \binom{(4)}{\kappa_1} p^2 + (266 \binom{(4)}{\kappa_{16}} + 53 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})) p^4) + 6 \binom{(4)}{\kappa_{15}} (2598 \binom{(4)}{\kappa_1} + 8 (163 \binom{(4)}{\kappa_{16}} + 21 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})) \binom{(4)}{\kappa_1} p^2 + \\
& (532 \binom{(4)}{\kappa_{16}}^2 + 212 \binom{(4)}{\kappa_{16}} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) + 11 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 p^4)) + 36 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2 \\
& (50176 \binom{(4)}{\kappa_{15}}^3 p^4 + 6272 \binom{(4)}{\kappa_{16}}^3 p^4 + 27 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 p^2 (20 \binom{(2)}{\kappa_1} + 3 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) p^2) + 24 \binom{(4)}{\kappa_{16}}^2 \\
& (1240 \binom{(2)}{\kappa_1} p^2 + 273 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) p^4) + 18 \binom{(4)}{\kappa_{16}} (9080 \binom{(2)}{\kappa_1}^2 + 780 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) \binom{(2)}{\kappa_1} p^2 + 83 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 p^4) + \\
& 96 \binom{(4)}{\kappa_{15}}^2 (1240 \binom{(2)}{\kappa_1} p^2 + 7 (112 \binom{(4)}{\kappa_{16}} + 78 \binom{(4)}{\kappa_3} + 39 \binom{(4)}{\kappa_4}) p^4) + 12 \binom{(4)}{\kappa_{15}} (27240 \binom{(2)}{\kappa_1}^2 + \\
& 20 (496 \binom{(4)}{\kappa_{16}} + 117 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})) \binom{(2)}{\kappa_1} p^2 + (3136 \binom{(4)}{\kappa_{16}}^2 + 2184 \binom{(4)}{\kappa_{16}} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) + 249 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 p^4)) + \\
& (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}})^2 (280576 \binom{(4)}{\kappa_{15}}^4 p^4 + 17536 \binom{(4)}{\kappa_{16}}^4 p^4 + 81 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^4 p^4 + \\
& 108 \binom{(4)}{\kappa_{16}} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 p^2 (50 \binom{(2)}{\kappa_1} + 9 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) p^2) + 384 \binom{(4)}{\kappa_{16}}^3 (275 \binom{(2)}{\kappa_1} p^2 + 72 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) p^4) +
\end{aligned}$$

	1	$\frac{3}{(4)} \frac{\kappa_1^{(2)}}{\kappa_1^{(4)}}$	$\begin{aligned} & 72 \kappa_{16}^{(4)} (11950 \kappa_1^{(2)} + 1200 (2 \kappa_3 + \kappa_4) \kappa_1 p^2 + 167 (2 \kappa_3 + \kappa_4)^2 p^4) + \\ & 1024 \kappa_{15}^{(4)} (825 \kappa_1 p^2 + 4 (137 \kappa_{16}^{(4)} + 54 (2 \kappa_3 + \kappa_4) p^4) + 96 \kappa_{15}^{(4)} (35850 \kappa_1^2 + \\ & 1200 (11 \kappa_{16}^{(4)} + 6 \kappa_3 + 3 \kappa_4) \kappa_1 p^2 + (4384 \kappa_{16}^{(4)} + 3456 \kappa_{16}^{(4)} (2 \kappa_3 + \kappa_4) + 501 (2 \kappa_3 + \kappa_4)^2) p^4) + \\ & 8 \kappa_{15}^{(4)} (17536 \kappa_{16}^{(4)} p^4 + 27 (2 \kappa_3 + \kappa_4)^2 p^2 (50 \kappa_1^2 + 9 (2 \kappa_3 + \kappa_4) p^2) + 288 \kappa_{16}^{(4)} (275 \kappa_1 p^2 + \\ & 72 (2 \kappa_3 + \kappa_4) p^4) + 36 \kappa_{16}^{(4)} (11950 \kappa_1^{(2)} + 1200 (2 \kappa_3 + \kappa_4) \kappa_1 p^2 + 167 (2 \kappa_3 + \kappa_4)^2 p^4)))))) / \\ & ((2 \kappa_{15} + \kappa_{16}) (2 \kappa_1 + 2 \kappa_{15} + \kappa_{16}) (9 \kappa_1 + 5 (2 \kappa_{15} + \kappa_{16}))^2 \kappa_1^{(2)}) \end{aligned}$
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		$\cdot \lambda_{15} + \lambda_{16}$	λ_{19} λ_{19}
			$(4)^2$ $(4)^2$ $(4)^2$ (4) (4) $(4)^2$ (4) (4)

[illegible]

Resolved unitarity condition(s):	(Demonstrably impossible)
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